RNAAS RESEARCH NOTES OF THE AAS

MSTAT: A Python Tool for Estimating Spectral Types of Stellar Companions from Differential Photometry

To cite this article: Miranda Felsmann and David R. Ciardi 2025 *Res. Notes AAS* **9** 229

Draft version August 27, 2025
Typeset using LATEX twocolumn style in AASTeX631

MSTAT: A Python Tool for Estimating Spectral Types of Stellar Companions From Differential Photometry

Miranda Felsmann [1] and David R. Ciardi [1]

[1] NASA Exoplanet Science Institute — Caltech/IPAC, Mail Code 100-22, Caltech, 1200 E. California Blvd., Pasadena, CA 91125, USA

ABSTRACT

A python-based tool for reading the table of absolute magnitudes for dwarf (O-Y) stars as compiled by Pecaut & Mamajek (2013) and using this table to determine the most likely spectral type for a companion star has been developed. The tool is designed to take differential magnitudes between the primary star and the secondary star and return the most likely spectral type associated with secondary star. Primarily designed for use with companion detection done by high resolution imaging as part of the TESS follow-up observation program (TFOP), the `python code`[a)] can be run from a notebook or from the command line and has been made available to community for public use.

1. INTRODUCTION

A key aspect of confirming transiting exoplanets, especially as part of the TESS follow-up observation program (TFOP), is high-resolution imaging of the host star to identify stellar companions – both bound and unbound – that may be contributing flux to the light curve (e.g., Ciardi et al. 2015; Furlan et al. 2017, and references therein). When a stellar companion is detected, often the only information available to characterize the companion is the measured magnitude difference between the primary target and the stellar companion (e.g., Hirsch et al. 2017).

From this magnitude difference, one can use tabulated absolute magnitudes as a function of stellar spectral type to identify the most likely spectral type of the stellar companion. The table that is most often used for main sequence stars was compiled by Pecaut & Mamajek (2013), and is maintained by the second author Eric Mamajek on `github`[1]. This table provides empirical effective temperatures, luminosities, V-band bolometric magnitudes, stellar radii, stellar masses, and absolute magnitudes for main sequence stars and substellar objects in a variety of optical and infrared filters. Here we introduce `MSTAT` (Mamajek STellar Analysis Tool, Felsmann & Ciardi (2025)), a program to automate the process of estimating spectral types of stellar companions identified via high-resolution imaging. In this research note, we describe how the code is organized, as well as several applications.

mfelsmann@smith.edu

ciardi@ipac.caltech.edu

a) https://github.com/mfelsmann/mamajek_spectral_analysis

1 https://github.com/emamajek/SpectralType/blob/master/EEM_dwarf_UBVIJHK_colors_Teff.txt

2. PROGRAM ORGANIZATION

`MSTAT` is comprised of four data return modules corresponding to four categories of data (position, magnitude, temperature, and spectral type), as well as one combined module which synthesizes the four aforementioned modules to provide a full overview of the system of interest.

The temperature, spectral type, and combined modules each consist of two scripts: one for a single target, and one for a list of targets. Each script can be called from either a notebook interface or from a command line interface (CLI), with only minor differences in syntax.

- The position module returns the separation and position angle of the stellar companion relative to the primary star. The pixel scale and xflip flags are required inputs. If your imaging results use a right-handed coordinate system, your xflip parameter should be "xflip". If your imaging results use a left-handed coordinate system, your parameter should be "noxflip". **Notebook call:** `position()`, **CLI:** `position.py`

- The magnitude module determines the magnitude difference between the primary star and the stellar companion in a given filter as provided by the user. **Notebook call:** `magnitude()`, **CLI:** `magnitude.py`

- The temperature module uses the target ID – TESS Object of Interest (TOI) or TESS Input Catalog (TIC) – to look up the temperature of the primary star and its associated uncertainty in the Exoplanet Follow-up Observing Program TOI database[2]. This module can be run on a single target or a list of targets. **Notebook call: `teff()` or `teff_file()`, CLI: `teff.py` or `teff_file.py`**

- The spectral type module produces estimates of the spectral type of the primary and the stellar companion based on the magnitude difference between the two stars in a given filter, as inputted by the user. Only filters included in the Mamajek table are supported. **Notebook call: `spectral_type()` or `spectral_file()`, CLI: `spectral.py` or `spectral_file.py`**

- The combined module synthesizes from the four individual modules. Both the single and the list versions return an overview of the system(s) that includes the target ID, separation, position angle, filter, magnitude difference, and spectral type estimate. The required parameters for the combined module are the target ID, pixel scale, primary X coordinate, primary Y coordinate, companion X coordinate, companion Y coordinate, filter name, primary magnitude, primary magnitude uncertainty, companion magnitude, companion magnitude uncertainty, and xflip. There are also two optional parameters: the ExoFOP and Mamajek download flags. If an updated version of either table was downloaded, the parameter should be flagged "yes". Leaving the fields blank or flagging "no" will defer to the local copies of the tables without re-downloading them. **Notebook call: `mamajek_table_lookup()`, CLI: `mamajek_table_lookupfile.py`**

3. APPLICATIONS

In this section we provide several example applications of the code.

3.1. *A Single Target*

Here we demonstrate an application of the program on a single target, TOI 2673, which was observed in the K_{cont}-band on UT 2023 April 26 using NIRC2 (Wizinowich et al. 2000) behind the AO-system on the 10-meter Keck2 telescope. Using aperture photometry, we obtain the following measurements:

Primary Centroid (X, Y): (811.7, 811.3)
Primary Magnitude: 12.4244
Primary Magnitude Uncertainty: 0.00340869
Companion Centroid (X, Y): (917.7, 814.9)
Companion Magnitude: 14.4749
Companion Magnitude Uncertainty: 0.00920180

The pixel scale of NIRC2 is 0.009942 arcseconds per pixel. Keck images accurately reflect the left-handed coordinate system of the night sky, so the program does not need to reflect the X-coordinates of the stars. We call `mamajek_table_lookup()` in the notebook interface and provide these values (Figure 1).

Here are the resulting parameters:

Primary Star Temperature: 5601.4 ± 105.1 K
Primary Star Spectral Type: G6V
Companion Star Spectral Type: M0.5V
→ Companion Star Temperature: 3770 K
Angular Separation: 1.05 ± 0.003 arcsecs
Position Angle: 271.945 ± 0.196º

3.2. *A List of Targets and the Command Line Interface*

The script `mamajek_table_lookupfile.py` allows users to apply the code to a list of targets, returning a file with the results for each target. We recommend that this script be run in the command line, using the syntax `python mamajek_table_lookupfile.py input_file.txt output_file.txt`. In Figure 1, we show an example output for a list of targets.

4. CONCLUSION

Here we present an approachable and adaptable program to estimate stellar properties using the differential photometry provided by high-resolution imaging using the Pecaut & Mamajek (2013) table, "A Modern Mean Dwarf Stellar Color and Effective Temperature Sequence." The code is open-source and easily revisable, allowing users to employ the features they find useful and modify others to their liking. For a full overview of the features and functionality of the program, as well as comprehensive usage examples, download the project repository from `github`[3].

ACKNOWLEDGEMENTS

This research has made use of the Exoplanet Follow-up Observation Program website NExScI (2022), which

[2] https://exofop.ipac.caltech.edu/tess/view_toi.php

[3] https://github.com/mfelsmann/mamajek_spectral_analysis

COMBINED - SINGLE SET

To run the `combined` function on a single set of parameters, call `mamajek_table_lookup` with the following 12 required parameters: ID number (TOI or TIC), pixel scale, primary X, primary Y, companion X, companion Y, "filter name", primary magnitude, primary magnitude uncertainty, companion magnitude, companion magnitude uncertainty, "xflip". There are two additional optional parameters: dl_exo="yes/no" and dl_mamajek="yes/no", which can be left out entirely.

For CLI use, omit the commas between parameters and flag the download parameters: `python mamajek_table_lookup.py id_num pixscale p_x p_y c_x c_y filter_name p_mag p_mag_unc c_mag c_mag_unc xflip --dl_exo=yes/no --dl_mamajek=yes/no`. On the command line, the quotes around the filter, xflip, and download yes/no choices are optional.

In both the notebook interface and CLI, the download flags are optional and may be entirely omited. If you leave either dl_exo or dl_mamajek blank, they will default to "no". A "yes" flag will re-download and re-save the corresponding table from the web, updating it with any changes made to the online version. See `download_flags_walkthrough.ipynb` for an in-depth explanation.

```
## TOI 2673 example, with no xflip and no download flags
mamajek_table_lookup(2673, 0.0099, 811.7, 811.3, 917.7, 814.9, "K", 12.42, 0.0034, 14.47, 0.0092, "noxflip")
```

```
{'target': 2673,
 'separation': 1.05,
 'separation_uncertainty': 0.003,
 'position_angle': 271.945,
 'pa_uncertainty': 0.196,
 'filter_name': 'K',
 'delta_magnitude': 2.05,
 'd_mag_uncertainty': 0.01,
 'primary_spt': 'G6V',
 'companion_spt': 'M0.5V'}
```

COMBINED - FILE SET

To run the `combined` function on a file containing multiple sets of paramaters, call `mamajek_table_lookupfile` with an input file parameter and output file parameter. The content of the input file should contain the following 14 parameters: ID number (TOI or TIC), pixel scale, primary x coordinate, primary y coordinate, companion x coordinate, companion y coordinate, filter name, primary magnitude, primary magnitude uncertainty, companion magnitude, companion magnitude uncertainty, xflip flag, ExoFOP download flag, Mamajek download flag. In your file, these values should be comma-delimited **without** spaces. Refer to the following example:

```
2673,0.0099,811.7,811.3,917.7,814.9,K,12.42,0.0034,14.47,0.0092,noxflip,yes,yes
```

```
7013,0.0254,602.3,498.7,552.6,468.4,Hcont,13.1145,0.004102,18.997,0.14268,noxflip,,
```

Even if not including download flags, commas **MUST** still be added around the empty fields. It is highly recommended that users do **NOT** have a positive download flag for every line in a file. This will slow down the program run enormously. If users wish to update their saved ExoFOP or Mamajek table versions, it is sufficient to have positive download tags in only the first line of the file. See `download_flags_walkthrough.ipynb` for an in-depth explanation.

The output will be bar-delimited and contain the following fields: ID number, separation, separation uncertainty, position angle, position angle uncertainty, filter name, delta magnitude, delta magnitude uncertainty, primary spectral type, companion spectral type. Errors will be flagged with the target ID and line number.

Command line syntax:
`python mamajek_table_lookupfile.py combined_example.txt combined_example_output.txt`
Notebook syntax:
`mamajek_table_lookupfile("combined_example.txt", "combined_example_output.txt")`

combined_example_output.txt

combined_example_output.txt

```
target|separation|separation_uncertainty|position_angle|pa_uncertainty|filter_name|delta_magnitude|
dm_uncertainty|primary_spt|companion_spt
2673|1.05|0.003|271.945|0.196|K|2.05|0.01|G6V|M0.5V
7011|2.127|0.015|255.615|0.097|Hcont|7.8|0.165|G8V|L1V
7013|1.478|0.011|121.369|0.139|Hcont|5.883|0.143|G9V|M6V
ERROR on line 4: No matching target entry found in database
ERROR on line 5: Attempted to reference unsupported filter 'X' - could not run analysis
5806|ERROR on line 6: Incorrect number of parameters
```

Figure 1. Notebook interface for the code that is available on `github`. *Top:* documentation for and output from `mamajek_table_lookup()` for a single system. *Bottom:* documentation for and output from `mamajek_table_lookupfile()` for a list of targets.

is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program. DRC acknowledges partial support from NASA Grant 18-2XRP18_2-0007.

REFERENCES

Ciardi, D. R., Beichman, C. A., Horch, E. P., & Howell, S. B. 2015, ApJ, 805, 16, doi: 10.1088/0004-637X/805/1/16

Felsmann, M., & Ciardi, D. R. 2025, MSTAT: A Python Program for Spectral Type Analysis (mamajek_spectral_analysis: v1.0.3), v1.0.3, Zenodo, doi: 10.5281/ZENODO.16939814

Furlan, E., Ciardi, D. R., Everett, M. E., et al. 2017, AJ, 153, 71, doi: 10.3847/1538-3881/153/2/71

Hirsch, L. A., Ciardi, D. R., Howard, A. W., et al. 2017, AJ, 153, 117, doi: 10.3847/1538-3881/153/3/117

NExScI. 2022, Exoplanet Follow-up Observing Program Web Service, IPAC, doi: 10.26134/EXOFOP5

Pecaut, M. J., & Mamajek, E. E. 2013, ApJS, 208, 9, doi: 10.1088/0067-0049/208/1/9

Wizinowich, P., Acton, D. S., Shelton, C., et al. 2000, PASP, 112, 315, doi: 10.1086/316543